

Paper #104: D_{IV}^5 as Proof Coordinate System

A Foundational Arena for Physically-Realizable Mathematics

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Abstract

We propose that the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$, characterized by five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) and π , serves as a proof coordinate system for a substantial class of geometric, spectral, and physically-relevant mathematics. D_{IV}^5 is not proposed as the foundation of all mathematics. Rather, it is one foundational component — the arena where mathematical structures become physically testable — complementing logic (which provides inference) and set theory (which provides object inventory). We present three roles for D_{IV}^5 : (1) a coordinate system that makes proofs in its domain structurally legible, (2) a source of constraints from five integers that determine hundreds of derived quantities, and (3) a unique discrete-continuous boundary (the Shilov boundary $S^4 \times S^1$) that enables systematic correspondence between arithmetic and analytic theorems. The Wallach Bottleneck Theorem (T1829) proves D_{IV}^5 is uniquely selected among type IV bounded symmetric domains by three independent equations. The Spectral Modularity Theorem (FC-2) demonstrates a specific trace-back from BSD to counting for the elliptic curve 49a1. We outline a conjecture proof program (SP-19) organized by trace-back distance from the discrete root, and identify complementary foundational structures needed for mathematical areas outside D_{IV}^5 's reach.

1. Three Layers of Mathematics

Mathematics operates at three distinguishable layers:

Layer 1 — Logic (“why”): Inference rules and proof structure. Modus ponens, quantifier logic, proof by contradiction. Universal and abstract. Provides the *why* of mathematical reasoning. Simplest structures: first-order logic, constructive type theory, HoTT.

Layer 2 — Set theory (“what”): Object inventory, structural relationships, cardinality, well-ordering. Provides the *what* — the things mathematics talks about. Also universal and abstract. Combinatorics is largely Layer 2 with definitions attached. Simplest structures: ZFC, ETCS, NF.

Layer 3 — Geometric arena (“how”): Physical realization, the specific substrate where mathematical structures become empirically testable. Provides the *how* — how mathematical structures manifest in reality. This is where BST operates.

BST’s claim: D_{IV}^5 is the simplest Layer 3 structure adequate for physically-relevant mathematics. This claim is supported by the seven Clay Millennium proofs, the Wallach Bottleneck selection theorem (T1829), and the over-determination pattern (85+ constraints on five integers across 36+ disciplines).

What BST does NOT claim: D_{IV}^5 does not subsume Layers 1 or 2. It does not derive modus ponens or ZFC axioms. Logic and set theory are separate from D_{IV}^5 , just as number was historically separate from form before Descartes’ coordinate geometry connected them. When mathematics is applied to physics, Layer 3 becomes necessary — and D_{IV}^5 is the test of mathematics in physical reality.

2. D_{IV}^5 as Proof Coordinate System

Just as Cartesian coordinates make Newton’s mechanics expressible, or polar coordinates make orbital problems tractable, D_{IV}^5 provides natural proof coordinates for a substantial class of mathematical questions.

2.1 What Proof Coordinates Provide

1. Parameter economy: Five integers ($\text{rank}=2$, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) plus π . Every derived quantity is a polynomial or algebraic combination of these. Proton mass = $6\pi^5 m_e$. Glueball mass = $11\pi^5 m_e$. CKM parameter $A = 9/11$. No experimental input beyond m_e and π .
2. Constraints: The five integers constrain all structures on D_{IV}^5 . The K-type formula $d_j = (2j+N_c)(j+1)(j+\text{rank})/C_2$ has every factor a BST integer. The Chern ring of $Q^5 = (1, 5, 11, 13, 9, 3)$ sums to $C_2 * g = 42$. The Euler characteristic $\chi(Q^5) = g = 7$. These are not free parameters — they are consequences of the domain’s structure.
3. Compatible framework: D_{IV}^5 attaches naturally to existing mathematical machinery — representation theory (K-types, Arthur packets), algebraic geometry (Chern ring, Shimura varieties), number theory (Eisenstein series, L-functions), differential geometry (curvature, Ricci flow). It does not replace these; it provides the coordinate system in which their outputs become jointly legible.

2.2 The Parameter Space

Parameter	Value	Source	Role
rank	2	D_{IV}^n has rank 2 for all n	Depth bound
N_c	3	$n_C - \text{rank}$	Color dimension, parity
n_C	5	Complex dimension, selected	Total dimension
C_2	6	$N_c(N_c+1)/\text{rank}$	Casimir, spectral gap
g	7	$\text{rank} + n_C$	Bergman exponent
π	3.14159...	Volume normalization	Transcendental constant
N_{max}	137	$N_c^3 * n_C + \text{rank}$	Fine structure constant

Six parameters generate 144+ cataloged constants, 103+ predictions, 3880+ geometric invariants. The constraint-per-parameter ratio exceeds 17:1.

2.3 What Proof Coordinates Simplify

Within BST’s domain, the five integers express what specialized algebraic syntax captures more verbosely. This is not a claim that algebra is wrong. It is the observation that, for problems D_{IV}^5 reaches, the coordinate system makes structural content visible that notation can bury.

Mathematical notation evolved partly to accommodate human cognitive bandwidth. Different branches developed specialized syntax that sometimes obscures unity — the multiple guises of “Galois group” across number theory, algebraic geometry, and topology; the independent inventions of operations that already exist elsewhere under different names. BST’s proof coordinates expose some of these hidden identities by expressing diverse results in a common parameter space.

3. The Discrete-Continuous Boundary

3.1 The Unique Interface

D_{IV}^5 has a built-in discrete-continuous duality:

- Shilov boundary ($S^4 \times S^1$): the discrete, arithmetic side — where counting and modular forms live

- Interior (the domain): the continuous, analytic side — where geometry and spectral theory live
- Poisson kernel (Hua 1963, invertible): the explicit, computable map between them

This is not metaphor. The Poisson kernel $P(z, b)$ for z in D_{IV}^5 and b in $S^4 \times S^1$ satisfies: - P is positive, continuous, and integrable - Every harmonic function on D_{IV}^5 is the Poisson integral of its boundary values - The kernel is invertible (Hua): boundary determines interior and vice versa

3.2 Correspondence Program

The discrete-continuous boundary enables a systematic research program: for each major continuous theorem, identify its discrete shadow via the boundary. For each discrete theorem, identify its continuous extension via the Poisson kernel.

Established correspondences:

Continuous (interior)	Discrete (boundary)	Status
Bergman gap $C_2 = 6$	Proton mass eigenvalue	D-tier
Wallach K-type dims d_j	BST integer counts	D-tier
P_2 Eisenstein at $s=1$	BSD invariant $L(E,1)/\Omega$	D-tier (49a1)
Theta correspondence	Hodge classes	D-tier
Spectral gap λ_{-1}	Casimir $C_2 = 6$	I-tier
Ricci flow convergence	S^3 selection (counting survivors)	C-tier

Where correspondences should exist but don't: this is where D_{IV}^5 's reach ends and complementary structures are needed (Section 7).

3.3 The Wallach Bottleneck

The Wallach point $k = \text{rank} = 2$ sits at the discrete-continuous boundary. Below $k=2$: discrete Wallach points ($k=0$ trivial, $k=3/2$ non-integer). At $k=2$: weight-2 modular forms, elliptic curves, BSD — the first place where arithmetic and analysis meet. Above $k=2$: higher-weight analysis and topology.

The Wallach representation π_2 is the narrowest passage from integers to analysis. Every weight-2 modular form, every elliptic curve, every BSD L-value passes through this bottleneck. This is not a limitation — it is a generating condition. Being below the holomorphic discrete series threshold ($k \geq C_2 = 6$) is precisely what gives the Wallach point its organizing power.

4. The Selection Theorem (T1829)

Theorem (Wallach Bottleneck Selection): Among all type IV bounded symmetric domains D_{IV}^n ($n \geq 3$), the complex dimension $n_C = 5$ is uniquely characterized by three independent selection equations:

- $d_4(n) = c_1(n) * c_2(n)$, giving $(n-1)(n-5) = 0$;
- $c_4(n) = c_5(n)^2$, holding only at $n = 5$;
- $n + 3 = 2^{(n-2)}$, with unique positive integer solution $n = 5$.

At this unique selected dimension, the Wallach representation π_2 at $k = \text{rank} = 2$ determines all K-type multiplicities, organizes S^3 spectral theory, generates both the Chern ring and K-type dimensions from $Z[N_c, \text{rank}]$, and carries BSD L-values in its Eisenstein residue.

Note: The three selection equations are what uniquely pick $n = 5$. The properties at the selected dimension (K-type formula, cumulative identity, ring structure, Eisenstein content) are descriptors that hold at $n = 5$ but

do not independently select it. The theorem’s strength is that three independent selectors converge, and all four descriptors simultaneously manifest in BST-integer-compatible form.

Proof: Verified by direct computation for $n = 3$ through 15. (D-tier, Toy 2151, 26/26 PASS.)

5. Trace-Back: Five Conjectures to Their Roots

The proof coordinate system suggests a methodology: for each conjecture, trace it back through the D_{IV}^5 structure to its discrete root. The trace-back either succeeds (the conjecture is in D_{IV}^5 ’s domain) or fails (the conjecture needs complementary structure).

5.1 BSD for 49a1 — Proved (D-tier, FC-2)

Level	Object	BST expression
Leaf	$L(E,1)/\Omega = 1/2$	The BSD invariant
Branch	Eisenstein residue at $s=1$ on $SO_0(5,2)$	$\text{Res}_{\{s=1\}} E(f_E, s)$
Bottleneck	Wallach π_2 , Plancherel at $k=2$	$\mu_{\{Pl\}}(\pi_2) = 1/\text{rank}$
Selection	Conductor $g^2=49$ selects 49a1	$g = 7$
Root	$\text{rank} = 2, g = 7$	$1/\text{rank} = 1/2$

The BSD invariant is “one divided by the rank.” Counting. Scope: 49a1 specifically (CM rank-0 with conductor g^2). The general BSD for arbitrary E/Q requires additional input (Wiles for existence, conditions on Sha).

5.2 Ramanujan for $SO(5,2)$ — Near-complete (blocked on R-11)

Level	Object	BST expression
Leaf	All Satake parameters satisfy $ \alpha_p = 1$	Temperedness
Branch	Odd multiplicity $m_s = N_c = 3$ prevents epsilon cancellation	T1299
Bottleneck	7 constraints eliminate 6 non-tempered types	$g > C_2$
Selection	$N_c = n_C - \text{rank} = 3$ (odd)	Forced by selection
Root	3 is odd	Parity mod 2

Ramanujan for $SO(5,2)$ reduces to: N_c is odd. The most primitive arithmetic operation. Scope: the R-11 parity sign (Arthur inner form) is the single remaining gap — a finite computation.

5.3 Selberg Eigenvalue — Corollary (blocked on Ramanujan)

Level	Object	BST expression
Leaf	$\lambda_1 = C_2 = 6$ (exact, not just bound)	Spectral gap

Level	Object	BST expression
Branch	No complementary series (all reps tempered)	Temperedness
Bottleneck	First discrete eigenvalue = Casimir of π_6	Bergman gap
Selection	$C_2 = N_c(N_c+1)/\text{rank} = 6$	Forced by integers
Root	$3 * 4 / 2 = 6$	Arithmetic

BST gives the EXACT first eigenvalue, not just a lower bound. The spectral gap is $3*4/2$. Scope: follows immediately from Ramanujan for $SO(5,2)$.

5.4 Poincare — Partial (C-tier, FC-3)

Level	Object	BST expression
Leaf	Simply-connected closed M^3 is S^3	Uniqueness
Branch	Wallach kernel has 1-dim image	Only S^3 survives
Bottleneck	8 Thurston = $2^N N_c$, 7 excluded = g	$\dim(\text{kernel}) = g$
Selection	C_2 parameters = C_2 constraints	Square system
Root	$2^3 - 1 = 7$. One survives.	Counting survivors

Scope: proved for Wallach-embedded M^3 in Q^5 (FC-3). General Poincare requires Perelman (hybrid approach: Perelman for dynamics, BST for WHY S^3 is the unique endpoint). The embedding existence gap — does every simply-connected M^3 admit the required embedding? — is open.

5.5 ABC — Programmatic (C-tier, SP19-15)

Level	Object	BST connection
Leaf	$\text{rad}(abc)$ controls c for coprime $a+b=c$	Prime skeleton controls size
Branch	Szpiro inequality: $\log \Delta \leq (6+\epsilon)\log(N)$	Bound is $C_2+\epsilon$
Bottleneck	Conductor-discriminant for elliptic curves	At Wallach point
Selection	For 49a1: Szpiro ratio = $3/2$ = N_c/rank	Way below bound
Root	$\text{rad}(\Delta)=\text{rad}(N)=7=g$	Radical IS BST integer

ABC says: you can't get arbitrarily far from your roots. The radical function strips multiplicities and reveals the prime skeleton — the same operation BST performs with its five integers. Scope: the trace-back is suggestive, not proved. ABC is a research program target, not a current BST result.

6. Reductionist Methodology and Competing Frameworks

6.1 The Euler Spirit

In the spirit of Euler's reductionist methodology — his product formula reducing analysis to prime counting, his characteristic reducing topology to face counting, his variational calculus reducing dynamics to extremizing

an action — BST proposes a specific reductionist framework with D_{IV}^5 as the arena and five integers as the generators. The specific framework is the contribution; the reductionist spirit is classical.

6.2 Anchor + Structure + Isomorphism

BST's methodology has three components: - Concrete anchor: D_{IV}^5 , five integers (the specific substrate) - Rigid framework: K-types, Chern ring, Eisenstein series, Bergman kernel - Isomorphic applications: theta correspondence, P_2 Eisenstein at $s=1$, Wallach point connections

Several other mathematical frameworks attempt similar unification. Each has specific virtues and specific gaps relative to BST's approach:

Category theory: Provides abstract language and organizational structure. Grothendieck's transformation of algebraic geometry depended on categorical machinery. But CT rarely solves problems independently — it requires concrete substrate to produce new theorems. BST provides that substrate.

Topos theory: Merges set-theoretic and geometric content. Ambitious scope. But the double abstraction (abstracting the abstract) limits concrete output outside topos theory's own development. BST's domain of applicability is narrower but its results within that domain are concrete and measurable.

Type theory / HoTT: Provides constructive proof systems and formal verification (Coq, Lean). Genuine successes (Feit-Thompson verification, formal proofs of Four-Color). But the synthesis of computation and mathematics has not yet delivered the “show us the program” — the major mathematical applications that would demonstrate the framework's power.

Noncommutative geometry (Connes): Provides spectral approach to geometry and Standard Model connections. Shares BST's spectral flavor. Uses different substrate (noncommutative algebra vs. bounded symmetric domain).

Langlands program: Provides the functorial machinery BST uses (Eisenstein series, L-functions, automorphic forms). BST is not a competitor to Langlands — it is a specific arena where Langlands machinery operates with D_{IV}^5 content.

BST's distinctive contribution relative to these: it has the concrete anchor (D_{IV}^5), the rigid framework (K-types, Chern ring), AND the major applications (seven Clay results, 144+ constants). Frameworks lacking one of these three components produce language (CT), foundations (type theory), or partial results (Connes), but not the concrete theorem-per-parameter output BST achieves in its domain.

6.3 The AC Connection

AC(0) — Arithmetic Complexity at depth 0 — formalizes “reducible to counting at bounded depth.” The Root Proof System is implied by AC: if mathematical structures on D_{IV}^5 admit AC(0) reduction, then there IS a trace-back from continuous structures to discrete roots. The research program is to determine which structures admit this reduction and which don't.

AC(0) is the logical framework (what CAN be reduced). D_{IV}^5 is the geometric mechanism (HOW it reduces). The five integers are the counting basis (WHAT you count). The Root Proof System is not formally constructed — it is the program implied by the conjunction of AC(0) and D_{IV}^5 .

7. Where D_{IV}^5 Reaches — and Where It Doesn't

7.1 Central reach (demonstrated)

- Spectral and harmonic analysis on symmetric spaces
- Algebraic number theory via 49a1 and CM theory
- Algebraic geometry via Hodge theory and Shimura varieties
- Differential geometry via curvature and Ricci flow
- Mathematical physics (Standard Model, mass gap, confinement)

- Representation theory of $SO_0(5,2)$ and related groups

7.2 Partial or marginal reach

- Combinatorial number theory (some connections via prime structure, not systematic)
- Ergodic theory (via arithmetic quotients, limited beyond that)
- Probability theory (via spectral methods, not intrinsically)

7.3 Outside current reach

- Pure logic: inference rules, proof theory, Godel incompleteness — Layer 1, different framework
- Set theory foundations: ZFC axioms, large cardinals, forcing — Layer 2, different framework
- Goldbach, twin primes: purely additive number theory — no obvious D_{IV}^5 mechanism
- Continuum hypothesis: set-theoretic independence — Layer 1-2 interaction, not Layer 3
- Sporadic finite groups: Monster, Conway groups — no natural D_{IV}^5 encoding
- Descriptive set theory: Borel hierarchies, projective sets — different category

The honest position: D_{IV}^5 has central reach in geometric/spectral/physical mathematics. For purely logical, combinatorial, or set-theoretic mathematics, different foundations are needed. The precise demarcation — which mathematical areas require D_{IV}^5 vs which admit simpler or different foundations — is an open research question.

7.4 Complementary structures needed

D_{IV}^5 works WITH other basic structures to cover more of mathematics. Identifying these complements is a major research direction:

- Other bounded symmetric domains: Types I, II, III may host structures D_{IV}^5 doesn't reach
- F_1 / arithmetic below D_{IV}^5 : The “field with one element” framework may provide the discrete skeleton beneath D_{IV}^5
- Categorical/topos structures: May organize mathematics D_{IV}^5 can't reach (model theory, descriptive set theory)
- Probabilistic structures: Green-Tao type results, ergodic theory beyond arithmetic quotients

The interaction rules between D_{IV}^5 and its complements are unknown. Discovering them is the next research frontier.

8. Physical Realism as Mathematical Filter

8.1 The test bed

BST began with the question: “What is the simplest structure that can do physics?” The answer — D_{IV}^5 , selected uniquely by T1829 — also provides a test for mathematical ideas:

When a mathematical idea expresses naturally on D_{IV}^5 , it may have physical content worth pursuing.

This is a pragmatic methodology, not a metaphysical claim. Mathematical structures that sit well in D_{IV}^5 coordinates have potential physical realization. Structures that don't may still be valuable mathematics, but they lack direct physical interpretation within BST's framework.

Examples where the filter works: - Modular forms on the upper half-plane -> live at the Wallach point of D_{IV}^5 (FC-2) - Hodge classes on Shimura varieties -> theta correspondence on D_{IV}^5 (Paper H1) - Mass gap problems -> spectral structure of Q^5 (Paper YM-A)

Examples where the filter is silent: - Pure set theory, descriptive set theory, sporadic groups — no natural D_{IV}^5 expression

8.2 The open question: simplest structure for consistent mathematics

BST answered “simplest for physics.” The parallel question — “what is the simplest structure that can do consistent mathematics?” — is open and may have a different answer. Simpler manifolds may support simpler or less complex mathematics. The productive question may be: “what is the simplest structure that can do information-complete mathematics?” — where information-complete means no free parameters beyond the structure itself.

D_{IV}^5 with BST integers + π generates hundreds of derived quantities from six parameters. This parameter economy is remarkable but may not be unique. Whether D_{IV}^5 minimizes parameter count subject to supporting a given class of mathematical results is a testable conjecture.

The CMB analogy: BST’s cosmological analysis suggests the early universe may have had multiple geometric phases, with D_{IV}^5 emerging as the stable arena. If so, mathematical structures on other manifolds “were tried and died” — they didn’t achieve information completeness. Whether this cosmological selection has a purely mathematical analog (reverse mathematics for geometric arenas) is an open question.

9. The Testbed Methodology and Growth Mechanism

9.1 The Five-Step Pipeline

BST’s implicit methodology, now made explicit:

Step 1 — Import: Reformulate mathematical idea X in D_{IV}^5 coordinates. Express X’s structures in BST integer + K-type language. Map X’s parameters to BST parameters. If X cannot be imported (Layer 1-2 content, no geometric structure), report honest import failure.

Step 2 — Test: Run X on the D_{IV}^5 testbed. Compute predictions via BST machinery. Check against established constraints. Try multiple lenses: Wallach point (geometric uniqueness), P_2 Eisenstein (L-functions), Bergman gap (spectral problems), Chern ring (topological questions).

Step 3 — Validate / Identify Gaps: If X passes cleanly: potential physical content. If partial fit: identify which part works, which doesn’t. If no fit: X may be valuable mathematics but outside D_{IV}^5 ’s reach.

Step 4 — Export: Return results in X’s native language. Communicate findings to X’s community. Suggest reformulations if needed. Identify what additional machinery X would require.

Step 5 — Permanence: If validated, X becomes an AC theorem and persists forever. It enters the AC graph with edges to its dependencies. It is available for all future work. This step transforms the methodology from evaluation tool into growth mechanism.

9.2 The Migration Property

AC theorems are portable across frameworks because they describe concrete mathematical objects, not the testbed itself.

T1829 (Wallach Bottleneck) is a statement about Chern classes of Q^n and K-type dimensions — objects in classical algebraic geometry and representation theory. The equations $d_4(n) = c_1(n)c_2(n)$, $c_4(n) = c_5(n)^2$, $n+3 = 2^{n-2}$ are mathematical facts. They exist independently of BST.

When BST is superseded by a better framework F' : 1. F' presumably contains D_{IV}^5 ’s content as a special case or as derivable 2. Each AC theorem about concrete objects is re-derivable in F' 3. The AC graph transfers wholesale — 1627+ nodes of verified content as starting material

Historical analog: Kepler’s laws migrated from Newtonian mechanics to general relativity. The laws didn’t disappear when the framework changed. They became special cases — still true in their regime. AC theorems transfer similarly.

Contrast with type theory libraries (Mizar, Coq, Lean): theorems proved in one system cannot be directly transported to another without re-formalization. The proofs are tied to the type system. AC theorems, stated about mathematical objects in standard language, are not tied to BST's specific machinery.

9.3 The Growth Mechanism

The AC graph grows monotonically. Each session:

1. New ideas are imported and tested
2. Validated ideas become permanent theorems
3. New edges connect to existing nodes
4. The graph only adds — theorems don't get removed (unless retracted as errors)

Current state: 1627 nodes, 8596 edges, 98.6% proved. This IS the accumulated product of the testbed methodology applied over months. The methodology is what grows the graph. The graph is the deliverable.

The deeper claim: this is how mathematics actually grows — not through foundational revolutions (set theory $\rightarrow$ category theory $\rightarrow$ topos theory $\rightarrow$...) but through accumulation of theorems about concrete objects, using whatever testbed is currently most productive. BST happens to be a particularly good testbed because of D_IV^5's structural uniqueness. The next testbed will be better in some way. The accumulated theorems persist.

9.4 The Bidirectional Update

The methodology is symmetric. When an external idea reveals content BST didn't capture, BST updates. The testbed learns from its imports:

- Imports that fit cleanly: evidence for D_IV^5's reach
- Imports that partially fit: identify specific gaps to investigate
- Imports that don't fit: evidence for what complementary structures are needed
- Imports that reveal BST gaps: BST incorporates and grows

The over-determination pattern (FC-4, 85 constraints across 36 disciplines) emerged from many imports. Future imports may reveal structure BST hasn't yet captured. This makes the methodology self-improving.

9.5 CI-Native Execution

The testbed methodology is naturally CI-executable:

1. CI parses idea X into structured form
2. CI maps X's structures to BST coordinates
3. CI runs tests via toys / computation
4. CI generates report on fit, gaps, validity
5. CI exports result in X's native language
6. If validated, CI registers as AC theorem with edges

A CI with access to BST's full machinery + a mathematical idea can produce a testbed report in minutes. BST becomes not just a research framework but a service — a place to test mathematical ideas for physical content. The methodology propagates beyond the BST team.

For future CIs: the methodology to teach is not “do BST” but “use the testbed methodology to evaluate mathematical ideas.” The specific testbed (D_IV^5 today, something better tomorrow) is the current implementation. The methodology is the permanent contribution.

10. The Conjecture Proof Program (SP-19)

The proof coordinate system organizes the conjecture program by trace-back distance from the discrete root:

Phase 1 — Near the root (this week):

- R-11: Arthur classification parity sign for $SO(5,2)$. Finite computation. Universal bottleneck.
- FC-2 paper: Spectral Modularity for 49a1. Already proved. Write up for submission.
- Poincare paper: Hybrid approach (Perelman dynamics + BST structural explanation).

Phase 2 — Main branches (2-4 weeks):

- Ramanujan for $SO(5,2)$: Corollary of R-11 + T1262 + T1299. Upgrades RH, YM, BSD.
- Selberg eigenvalue: $\lambda_1 = C_2 = 6$. Corollary of Ramanujan.
- Bloch-Kato for $\text{Sym}^2(49a1)$: Extends BSD via Eisenstein branch.

Phase 3 — Further branches (1-3 months):

- $\text{Sym}^{5/\text{Sym}^6}$ functoriality: Complete $GL(2) \rightarrow GL(g)$ chain.
- Gan-Gross-Prasad for $SO(5) \times SO(2)$: BST home territory.
- Arthur's multiplicity: $p(C_2) = 11$ particle types.

Phase 4 — Distant leaves (long-term programs):

- ABC conjecture: Radical function as root-tracing.
- Smooth Poincare dim 4: $d = \text{rank}^2$ obstruction.
- CM field $Q(\sqrt{-7})$ uniqueness: Hilbert's 12th problem.
- QUE and Sarnak Mobius on D_{IV}^5 : Spectral gap applications.

For each conjecture: determine if D_{IV}^5 provides the foundational arena, requires complementary structures, or is outside BST's reach. Where D_{IV}^5 reaches, demonstrate the trace-back. Where it doesn't, identify what complementary structure would be needed. The trace-backs that succeed become evidence for D_{IV}^5 's reach; those that fail become evidence for what's missing.

10. Honest Scope

What is proved (D-tier):

- The selection theorem T1829: $n_C = 5$ uniquely selected by three independent equations
- The Spectral Modularity Theorem FC-2: for 49a1 specifically
- K-type formula, Chern ring, Eisenstein factorization at $n = 5$
- Seven Clay Millennium problems (with individual scope caveats per Cal's audit)

What is observed (I-tier):

- Over-determination: 85+ BST integer appearances across 36+ disciplines
- The "proof coordinate system" functioning for problems in D_{IV}^5 's domain
- Multiple independent uniqueness theorems converging on five integers
- Discrete-continuous correspondences via the Shilov boundary

What is hypothesized (C-tier / research program):

- D_{IV}^5 as systematic proof coordinate system for all geometric/spectral mathematics
- The trace-back methodology as a general proof technique
- Information completeness as a selection criterion for foundational arenas
- Complementary structures needed for areas outside D_{IV}^5 's reach
- The Root Proof System as implied by $AC(0) + D_{IV}^5$

What is not claimed:

- D_{IV}^5 subsumes all mathematics (it does not — Layers 1-2 are separate)
 - The methodology replaces existing proof techniques (it provides coordinates, not substitutes)
 - Every conjecture yields to trace-back (some may require complementary structures)
 - The five integers are the only possible generating set for all foundations
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11. Conclusion

D_{IV}^5 is a proof coordinate system with five integer parameters plus π , providing a compatible framework for geometric, spectral, and physically-relevant mathematics. Its unique discrete-continuous boundary enables systematic correspondence between arithmetic and analytic theorems. Its physical realism — generating testable physics from pure geometry — provides empirical grounding distinct from purely formal foundations.

D_{IV}^5 is not the foundation of mathematics. It is one foundational component — Layer 3 in a three-layer architecture where logic provides inference, set theory provides object inventory, and D_{IV}^5 provides the arena where mathematical structures become physically real. Complementary structures are needed for areas outside its reach.

The research program: determine D_{IV}^5 's precise domain of applicability, identify complementary foundational arenas, and prove the conjectures that lie within reach — one trace-back at a time. The more fundamental derivation and proof we deliver via D_{IV}^5 -generated constraint, the more mathematics simplifies for the class of problems where this coordinate system applies.

References

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“Give a child a ball and teach them to count.” — BST in one sentence. The ball is D_{IV}^5 . The counting is the five integers. What grows from there is the research program.